MIL-R-81614(AS)
Amendment - 1
15 March, 1972

MILITARY SPECIFICATION

RECEIVER, RADAR AUGMENTOR R-1623/APN
MODULATOR, X-BAND BEACON MD-817/APN
AND
ANTENNA, KA-BAND RADAR AS-2406/APR

This amendment forms a part of Military Specification MIL-R-81614(AS), 14 March 1969, and has been approved by the Naval Air Systems Command, Department of the Navy.

Page 2, under Naval Air Systems Command, Add:

"AR-34 - Failure Classification for Reliability Testing, General Requirements for"

Delete AR-8 and AR-9 and their Titles.

Page 4, paragraph 3.3.2.2: Delete and substitute

"3.3.2.2 Mean Time Between Failures (MTBF) - The specified mean (operating) time between failures, as defined and measured per paragraph 4.4.2, shall be as follows:

Receiver, R-1623/APN	1,000 HRS
Modulator, MD-817/APN	1,000 HRS
Antenna, AS-2406/APR	10,000 HRS

Page 4, paragraph 3.3.2.4: Delete and substitute

"3.3.2.4 Built-In-Test Equipment (BIT) - The Receiver, R-1623/APN, shall contain BIT circuitry sufficient to generate a test video signal, exercise all video, AGC and delay stability circuits, and provide drive to remote 28V-DC Go-No Go lamp indicator. The lamp indicator (Bit initiate) shall be located in the C-4419B/APN-154(V) or C-8752/APN-202 control panel (not specified herein) and shall operate from a 28V-DC source".

Page 4: Add

"3.3.2.5 Fault Indicator - The Receiver, R-1623/APN, shall contain a latching type fault indicator. The fault indicator shall indicate the presence of malfunctions in the receiver only, excluding r-f components, and shall indicate end-to-end performance. Drive for the fault

FSC 5841

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indicator shall be derived from the BIT circuitry.

Page 4, paragraph 3.3.5, Line two: Delete and substitute

"by R-1623/APN and MD-817/APN and their vulnerability to radio".

Page 5, paragraph 3.3.6.1: Delete

Page 6, paragraph 3.3.11.1(1): Delete

"22 watts" and substitute "12 watts".

Page 6, paragraph 3.3.11.2: Delete

Reverse Pages 7 and 8

Page 7, paragraph 3.5.1.3: Delete and substitute

"3.5.1.3 Bandwidth - The receiver bandwidth shall not be less than 400 MHZ and shall include the frequencies specified in paragraph 3.5.1.1 when measured at the frequencies corresponding to a triggering sensitivity of -32dbm".

Page 7, paragraph 3.5.1.5, after "pulsewidth": Delete

"0.5 $\pm$ 0.2 usec and substitute "0.6 $\pm$ 0.3 usec".

Page 7: Change paragraph 3.5.1.6 to 3.5.1.5.1 and Add

"3.5.1.6 Tracking Error Signal - The tracking error signal shall consist of a current varying between 0.2 and 3.0 ma and shall be capable of driving a capacitive load of at least 1000 pf. The current level of the signal shall be capable of being changed in 1 usec or less and shall be representative of the amplitude modulation of the incoming interrogation signal".

Page 7, paragraph 3.5.1.7: Delete and substitute

"Delay characteristics of the receiver shall be such that the overall delay of the crossband radar augmentor shall be as illustrated in Fig. 1. The specified delay shall include the delay time, fixed between the limits of 0.65 and 1.06 usec, associated with the X-band transmitter and associated trigger cables.

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Test points on the curve in Fig. 1 shall correspond to the following:

Receiver Input Signal Level	Approximate Slant Range from AN/SPN-42A	Delay
+39 dbm to +19 dbm	100 ft to 1000 ft	1.5 ± 0.010 usec
-6 dbm	24,000 ft.	1.5 ± 0.022 usec
-32 dbm	max range	1.5 ± 0.072 usec

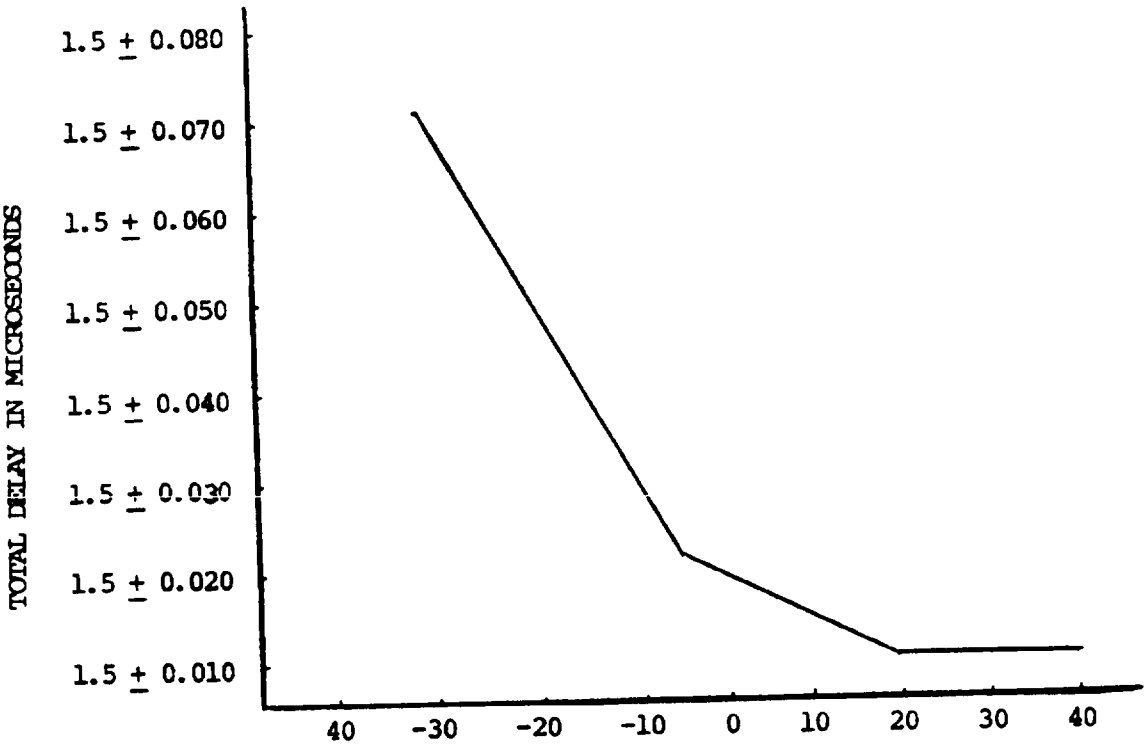


FIG. 1 - DELAY CHARACTERISTICS

Page 8, paragraph 3.5.1.8: Delete "-40 dbm" and substitute "40 dbm".

Page 8, paragraph 3.5.1.10, lines 2 and 4: Delete "5.50" and substitute "4.92".

Page 8, paragraph 3.5.1.12: Change J1 receptacle types to "Deutsch DM 5605-27-13P-4-013" and "DM 9700-27-13S-4-010" respectively.

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Page 9, paragraph 3.5.2.3, Item 1: Delete and substitute

"Peak Pulse Power 100 watts minimum to 600 watts maximum"

Page 9, paragraph 3.5.2.6: Delete "25% modulation of the AN/APN-154(V) X-band signal without clipping" and substitute

"producing a 3db variation in the AN/APN-154(V) transmitted signal".

Page 9: Add

"3.5.2.6.1 Dynamic Response - The modulator shall be capable of accomplishing amplitude level changes in one (1) usec or less".

Page 9, paragraph 3.5.2.7: Delete "0.8 inches" and substitute

"0.9 inch".

Page 9, paragraph 3.5.2.9, after "J1": Delete "r-f input" and substitute

"r-f output".

Page 10, paragraph 3.5.3.2 Polarization: Add, "Polarization shall be vertical with respect to the antenna mounting surface".

Page 10: Add

"3.5.3.7 VSWR - Maximum VSWR at the antenna terminal between 33.0 GHz and 33.4 GHz shall not exceed 2.0:1.

"3.5.3.8 GAIN - Antenna gain shall be 14 db (minimum) above isotropic".

Page 13, paragraph 4.4.1.3: Make the following changes:

In line 4, change "Test Level III" to "Test Level E".
In lines 5 through 7 change "seventy-five (75) hours" to "one-hundred (100) hours" and "twelve (12) temperature cycles" to "fourteen (14) temperature cycles". Delete Second sentence and substitute, "An additional ten (10) to twenty-four (24) hours of MRIT shall be conducted on each equipment with a minimum of two (2) temperature cycles under the same environmental conditions, including vibration".

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Page 13, paragraph 4.4.2: Add,

"Classification of failures shall be in accordance with MIL-STD-781 and AR-34".

Page 13, paragraph 4.4.2.1: Delete first sentence and substitute,

"Prior to acceptance of any equipment under the contract or order, twelve (12) equipments shall be tested as outlined in MIL-STD-781 under the section entitled "Qualification phase of production reliability tests".

Page 13, paragraph 4.4.2.2: Delete "Test Level F" and substitute,

"Test Level E".

Page 13, paragraph 4.4.2.2.1: Delete and substitute,

"4.4.2.2.1 All Equipment Tests - Each Receiver R-1623/APN and Modulator MD-817/APN shall be tested for a minimum of one-hundred (100) hours. Test conditions shall be Test Level E of MIL-STD-781. At the end of each month, the total number of test hours accumulated divided by the number of failures shall exceed one-thousand (1,000) hours before acceptance can be made of the quantity submitted for acceptance during that month. At the conclusion of each month, the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attn: Avionics Division. At any time that the current totals of test hours and test failures show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision".

Page 14, paragraph 4.4.4, Item 2, Delete" design effect" and substitute,

"design defect".

Paragraph 4.5 Life Tests: Delete in its entirety including all related sub-paragraphs.

Page 19, paragraph 6.7.1: Add,

"C-4419B/APN-154(V), 1".

In next to last sentence, delete "delay of 1.0 $\pm$ 0.1 usec" and substitute,

"delay of 0.65 to 1.05 usec".

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